

Growth Stages of Rice Crop

Source: Tara Group Pakistan

Introduction

Rice is a cereal belonging to the family Poaceae. As a tropical crop, it can be grown during the year's two distinct seasons, dry and wet, provided sufficient water is available. Rice is normally an annual crop, but in the tropics it can survive as a perennial, producing a ratoon crop.

Overview of Growth Stages

The growth stages of rice are germination, seedling, tillering, internode elongation, booting, heading, flowering, milking, dough, and maturation.

Germination

Seed germination occurs when the seed coat has absorbed adequate water to become soft and elastic. The ideal temperature for rice germination is 87°F, or 30°C.

Seedling

Seedling emergence occurs when the first internode, called the mesocotyl, has elongated and pushed the tip of the rice coleoptile, also called the epiblast, through the soil surface. The appearance of the coleoptile is a telltale sign of seedling emergence.

Tillering

Tillering usually begins at the fifth-leaf stage, referred to as V5, when the first tiller becomes visible and emerges from the axillary bud of the second leaf on the culm. Tillering continues synchronously with the n th leaf of the main culm, with each tiller emerging from the axillary bud of the $(n-3)$ th leaf. The first tiller that emerges is called the primary tiller. Secondary tillers grow from the primary tillers, and tertiary tillers are produced from the secondary tillers.

Internode Elongation

Internode elongation begins about the time panicle initiation is occurring, continues until full plant height is reached, and is followed by heading. The process of panicle differentiation in rice happens while internode and panicle growth is ongoing.

Booting

The booting stage is loosely defined as the period characterized by a swelling of the flag leaf sheath, which is caused by an increase in the size of the panicle as it grows up the leaf sheath. Booting is the stage in which meiosis occurs.

Heading

Heading, referred to as the R3 growth stage, is the time when the panicle begins to exert from the boot. Heading begins at about 60 days, depending on the variety grown. During this stage, the cells of the plant stem enlarge rapidly and vertically, pushing the panicle upwards. The stem becomes soft during this period and susceptible to attack by stem borer.

Flowering

Flowering refers to the events between the opening and closing of the spikelet, or floret, and lasts for 1 to 2½ hours, during which time pollination occurs. Flowering generally begins upon panicle exertion or on the following day, and is consequently considered synonymous with heading.

Ripening

The grain filling and ripening, or maturation, phase covers the R6 to R8 growth stages. It follows fertilization of the ovary and is characterized by grain growth. During this period, the grain increases in size and weight. The grain changes color from green to gold or straw color at maturity, and the leaves of the rice plant begin to senesce.

Milking

At the milking stage, the developing starch grains in the kernel are soft, and the interior of the kernel is filled with a white liquid resembling milk.

Dough

The dough stage has two phases. In the soft dough stage, the starch in the grain is beginning to become firm but is still soft. In the hard dough stage, which includes the end of grain filling (R7) and the grain-drying stage (R8), the whole grain is firm and almost ready for harvest.

Maturity

At maturity, the whole grain is hard and ready for harvest. This stage is reached at approximately 20 to 22 percent grain moisture.